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Microstructure and mechanical properties of a refractory $HfNbTiVSi_{0.5}$ high-entropy alloy composite

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ABSTRACT

A refractory high-entropy alloy of $HfNbTiVSi_{0.5}$ was synthesized by induction levitation melting. The alloy is a composite composed mainly of a body centered cubic (BCC) solid solution and a multi-component silicide ((Hf, Nb, Ti)-Si). This alloy displays a balanced combination of strength at high temperature and ductility at room temperature. The studied results show that the compressive yield strength (YS) is as high as 1399 MPa, with a fracture strain of 10.9% at room temperature. At 800 °C and 1000 °C, the YS is still preserved at 875 MPa and 240 MPa, respectively, due to the higher load-carrying capacity of the silicide at elevated temperatures. Comparison to similar refractory HEAs shows that, the composite microstructure is beneficial for better room temperature strength and ductility as well as the elevated temperature properties.

1. Introduction

High entropy alloys (HEAs) defined as those containing generally at least five principal elements have been designed according to the demands such as oxidation resistance, strength, ductility or other properties [1–3]. Recently, to develop refractory alloys for applications in advanced gas turbines or airplane engines, intense research effort has been devoted to HEAs with higher melting temperature based on elements such as Ta, Mo, W, Nb, Hf, Ti, V, and Zr [4–7]. Although the systems with Ta, Mo, W show good temperature capability, they have a disadvantage of high densities greater than 9.90 g/cm^3 [5–7]. In addition to the lower density, another excellent feature of the HEAs based on Nb, Hf, Ti, V, Zr, etc. is the remarkable room temperature ductility arising from the BCC phase solid solution [8–11]; however, these alloys start to soften at intermediate temperatures not higher than 800 °C [11]. Therefore, enhancement of their elevated temperature strength is highly important. In a novel approach, Cr was added to these alloys to generate Cr_2M Laves compound in-situ in order to strengthen the BCC solid solution matrix [10–13]. However, the strengthening effects of the Cr_2M Laves phase at ultra-high temperatures are limited due to the lower melting point. Moreover, the appearance of the Cr_2M Laves phase leads to a significant drop in room temperature ductility [10,11,13].

Considering that silicon can be combined with many metallic elements to form stable compounds that exhibit ultra-high melting points, higher temperature strength, and good oxidation resistance [14–16], this paper reports an attempt to fabricate HEAs-based composites through the addition of an appropriate content of silicon to generate silicides in-situ in order to strengthen the HEAs that demonstrate remarkable compressive ductility at room temperature [8,10,11,17]. Actually, this approach has been used previously for the fabrication of Nb-silicide-based ultra-refractory alloys [16]. Based on the above considerations, Nb, Hf, Ti, V, Si were selected. We hope that this approach will enable the achievement of a balanced combination of strength at high temperatures and ductility at room temperature. For this purpose, this paper systemically investigated the mechanical properties and fracture behavior of $HfNbTiVSi_{0.5}$ over a wide temperature range.

2. Experimental procedures

The alloy with nominal composition of $HfNbTiVSi_{0.5}$ was synthesized by cold crucible levitation melting in an argon atmosphere. Raw elements with purities higher than 99.9 wt% were used. The sample was re-melted two times to ensure chemical homogeneity.

Crystal structure was identified using D8-ADVANCE X-ray diffractometer. Microstructure and chemical composition were

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analyzed using Zeiss MERLIN-VP-COMPACT scanning electron microscope (SEM) equipped with energy-dispersive (EDS) detector. The three-dimensional morphology was investigated using OLS4100 laser microscope. Vickers microhardness measurements were conducted using MH-5L microhardness tester. Compression tests at room temperature, 800 °C, 1000 °C were conducted using Gleeble-3500D thermal-mechanical simulator under an initial strain rate of 0.001 s^{-1}. Cylindrical specimens for compression testing had a diameter of 6.0 mm and a height of 9 mm. All tests were performed until the fracture of the specimen or until a height reduction of 50% was reached.

3. Results and discussions

Representative X-ray diffraction pattern of $HfNbTiVSi_{0.5}$ alloy is presented in Fig. 1a. The diffraction peaks, as indicated in the figure, demonstrate the presence of multiphase structure: mixed hexagonal compound (JCPDS 29-1362) with lattice parameter a = 764.9 pm, c = 525.4 pm and disordered BCC phase (JCPDS 35-789) with lattice parameter of 326.7 pm.

SEM micrograph of the as-cast $HfNbTiVSi_{0.5}$ alloy shown in Fig. 1b reveals a composite structure with silicide phases distributed in the BCC matrix as designed. According to the EDS analysis (Table 1), the BCC matrix is enriched with Hf, Nb, Ti, V, and depleted of Si, while the silicide is rich in Hf, Si and depleted of V. Due to their lower enthalpy, Si addition results in the formation of compounds of Si with Hf, Nb, Ti, V ($\triangle H_{Hf\text{-}Si} = -77$ kJ/mol, $\triangle H_{Nb\text{-}Si} = -56$ kJ/mol, $\triangle H_{Ti\text{-}Si} = -66$ kJ/mol, $\triangle H_{V\text{-}Si} = -48$ kJ/mol [18]). Comparison of chemical compositions confirms that Hf has a stronger chemical affinity with Si element while little V is present in silicide. It is important to mention that a multi-component silicide solution ((Hf, Nb, Ti)-Si) rather than multiple-phases of silicides are formed in the alloy. This gives rise to a general expectation that the produced multi-component compound in HEAs exhibits combination properties. Additionally, a small amount of gray phase enriched with V is observed along the silicide boundaries. The appearance of the V-rich phase is possible due to the lower solubility of V in the matrix and the higher atomic radius difference between V and Hf (134 and 158 pm, respectively) [9].

Several parameters have been provided as a guideline for predicting the phase formation and the structure in HEAs. In the present study, formation of silicide and BCC matrix in $HfNbTiVSi_{0.5}$ is correctly predicted by the parameters $\delta = 9.0\%$, $\Omega = 1.25$ ($\delta \leq 6.6\%$ and $\Omega \geq 1.1$ for stabilized solid-solution phase, values in

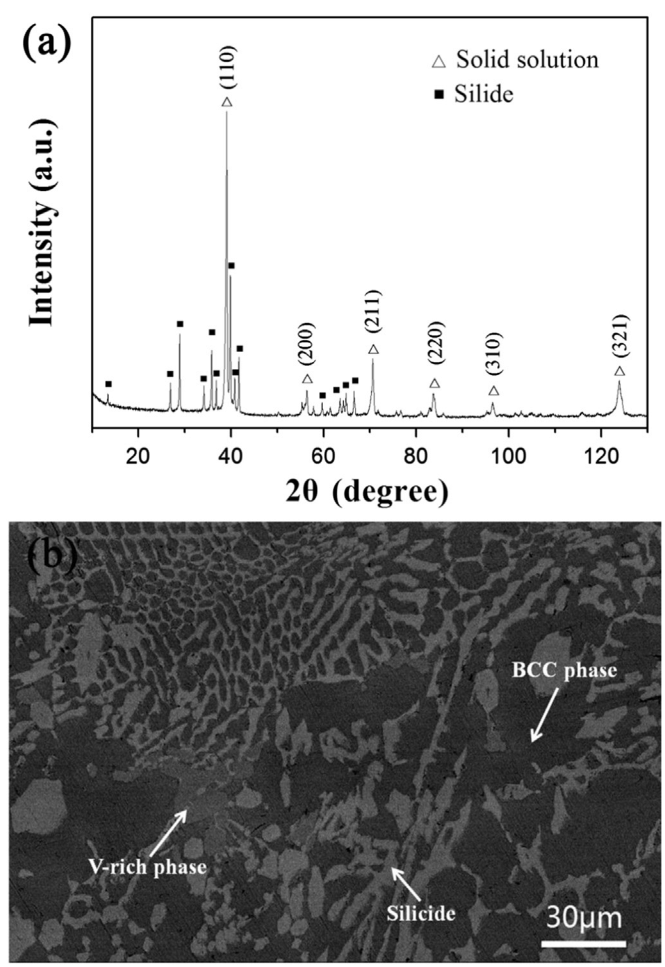

Fig. 1. Microstructure of the $HfNbTiVSi_{0.5}$ alloy: (a) XRD pattern; (b) SEM image.

Table 1
Chemical composition (in at%) of the $HfNbTiVSi_{0.5}$ constituent.

Element	Hf	Nb	Si	Ti	V
Nominal	22.2	22.2	11.2	22.2	22.2
Silicide phase	31.1	12.8	38.5	13.0	4.6
BCC phase	17.9	36.3	1.9	21.3	22.6
V-rich phase	30.7	9.8	8.4	10.6	40.5

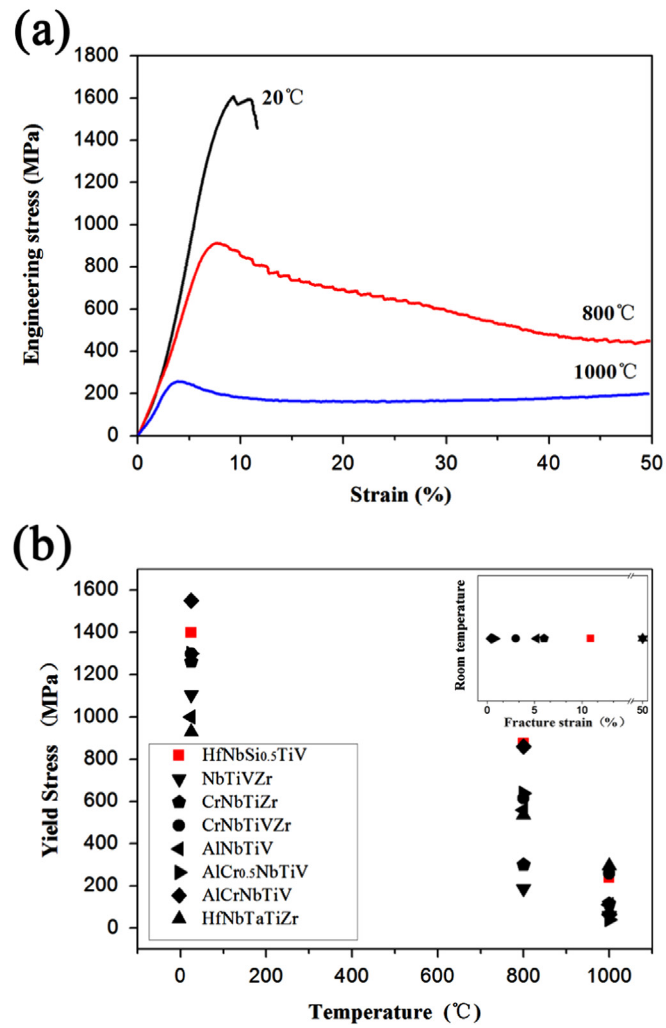

Fig. 2. Compression results and comparison: (a) stress-strain curves, (b) comparison of mechanical properties between $HfNbTiVSi_{0.5}$ and similar refractory alloys.

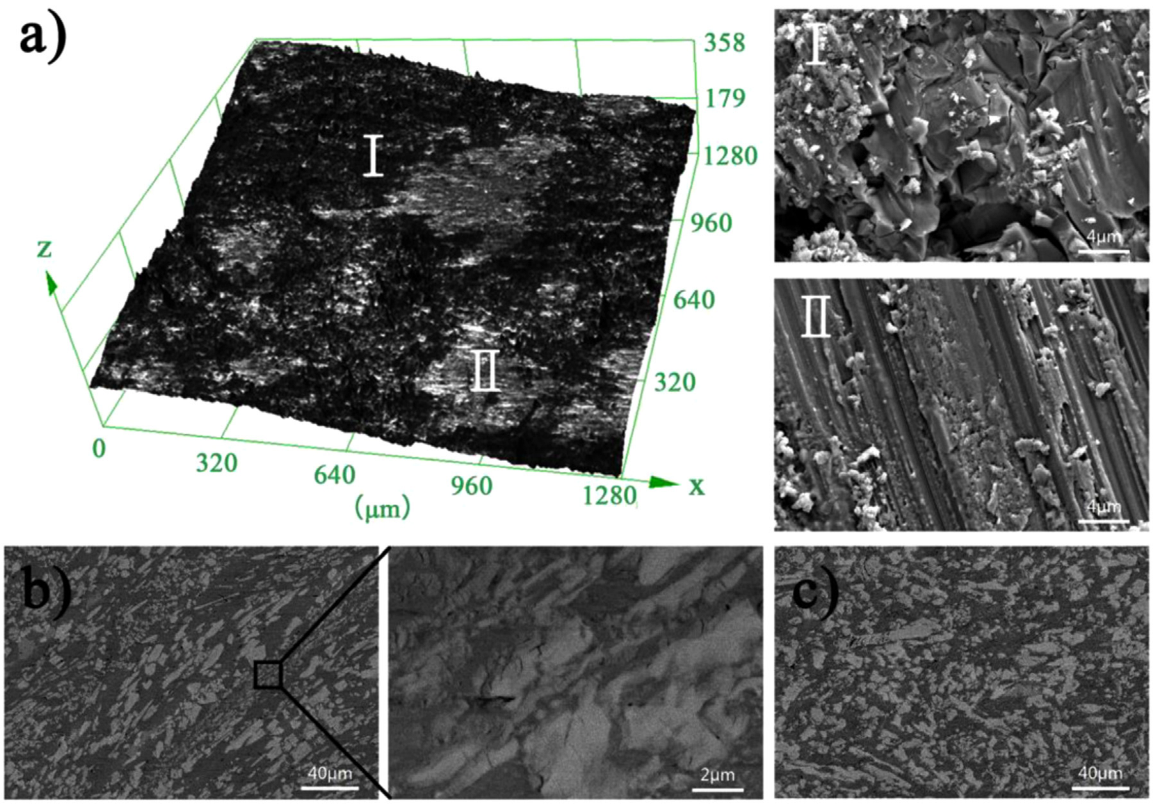

Fig. 3. 3D and SEM images of $HfNbTiVSi_{0.5}$ after compression (a) at room temperature, (b) at 800 °C, (c) at 1000 °C.

another range for ordered compounds or BMGs [19])and VEC=4.44 (VEC ≤ 6.87 for BCC structure [20]). It should be noted that the δ, Ω and VEC parameters are still applicable for predicting structural stability and phase formation though there were many silicides forming in $HfNbTiVSi_{0.5}$.

The measured density of $HfNbTiVSi_{0.5}$ is $\rho=8.60\ g/cm^3$. The measured Vickers microhardness values of BCC phase and silicide are 420 ± 12 HV and 756 ± 33 HV, respectively. The formation of silicides strengthens the matrix, leading to an overall hardness of 490 HV.

The compressive stress-strain curves of $HfNbTiVSi_{0.5}$ at different temperatures are presented in Fig. 2a. At room temperature, the $HfNbTiVSi_{0.5}$ alloy exhibits a high yield stress of 1399 MPa, a peak stress of 1608 MPa, and a fracture strain of 10.9%. Although a severe softening stage is observed from the elevated temperature deformation curves, the $HfNbTiVSi_{0.5}$ alloy shows attractive elevated temperature properties, with a yield stress of 875 MPa and 240 MPa at 800 °C and 1000 °C, respectively, with no brittle fracture occurring at a strain of 50%. The different compression performances at different temperatures depends on matrix behavior, properties of (Hf, Nb, Ti)-Si, and the solution hardening effect.

As shown in Fig. 2b, the $HfNbTiVSi_{0.5}$ alloy demonstrates ultra-high YS with remarkable ductility compared to similar refractory HEAs at room temperature and 800 °C. At 1000 °C, the $HfNbTiVSi_{0.5}$ alloy exhibits further perfect mechanical performance: stronger YS close to that of TaNbHfZrTi and four times higher than that of NbTiVZr [7,11]. Compared to the Laves phase-strengthen alloys, the YS of $HfNbTiVSi_{0.5}$ at elevated temperatures is also highly attractive, and stronger than the majority of reference alloys [11,13]. These results demonstrate that the addition of Si has beneficial effects on the elevated temperature strength as well as on the comprehensive mechanical properties.

Fig. 3a shows a typical fracture surface of $HfNbTiVSi_{0.5}$ fractured after 10.9% compression strain at room temperature. Microcracks are clearly observed in Fig. 3a (I), demonstrating typical cleavage fracture of the silicide during deformation. Broken silicides generate grooves along the glide plane of the matrix when plastic deformation occurs, and tend to adhere to the matrix to prevent further slippage. Instead, no noticeable cracks are initiated inside the matrix though the sample was heavily deformed. These results indicate a hybrid elastic-plastic fracture at room temperature. Evidently, the majority of the ductility is derived from the matrix, while the majority of the strength increase is attributed to the load-carrying capacity of silicides and the plowing effect.

Microstructures of $HfNbTiVSi_{0.5}$ after compression at 800 °C and 1000 °C are shown in Fig. 3b and c. Obvious deformation bands can be observed from the longitudinal sections, demonstrating that the matrix softened due to intense thermal activation effect [7]. The silicides are severely deformed by the effect of plastic flow, and become more refined and homogeneous than in the as-cast conditions. It is evident that the attractive strength at elevated temperatures is strongly dependent on the strengthening effect cause by the silicides. Several microcracks are observed inside the silicides, while no cracks between matrix and silicides can be seen in the magnified image. This demonstrates the high interfacial adhesion between the matrix and silicides.

4. Conclusions

To summarize, a new refractory $HfNbTiVSi_{0.5}$ alloy mainly composed of a BCC solid solution and a simple multi-component silicide has been successfully developed. The alloy density is 8.60 g/ cm^3 and the overall hardness is 490 HV. At room temperature, the YS is 1399 MPa, and the fracture strain is 10.9%. At 800 °C and 1000 °C, no fracture occurs at the strain of 50%, and YS values of 875 MPa and 240 MPa, respectively, are obtained.

Compared to similar refractory HEAs, the composite structure incorporating the silicide shows superior comprehensive strength and highly improved elevated temperature properties.

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